

Tidy data, joining and project organization

Data Science Flipped Classroom - Module 7

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Tidy data

The same data, many shapes

The same information can be stored in many different table layouts.

Most of them make analysis harder than it needs to be.

Tidy data is one consistent layout that fits the tidyverse, so `dplyr`, `ggplot2`, and the rest all work without reshaping first.

Data semantics

Before we can say what makes a dataset tidy, we need a shared vocabulary for what is in it. A **dataset** is a collection of values, each one a number or a string.

Every value belongs to both a variable and an observation:

- A **variable** holds all the values that measure the same attribute (height, temperature, duration) across units
- An **observation** holds all the values measured on the same unit (a person, a day, a country-year) across attributes

The three rules

- Each **variable** forms a column
- Each **observation** forms a row
- Each **cell** is a single measurement

If your table follows these three rules, it is tidy. Almost everything else this module does is about getting there.

Tidy in practice

`tidyr` ships with the same dataset in several shapes. This one is tidy:

```
1 table1
```

```
# A tibble: 6 × 4
  country      year  cases population
  <chr>      <dbl> <dbl>      <dbl>
1 Afghanistan 1999     745  19987071
2 Afghanistan 2000    2666  20595360
3 Brazil      1999   37737  172006362
4 Brazil      2000   80488  174504898
5 China       1999  212258  1272915272
6 China       2000  213766  1280428583
```

One row per country-year, one column per variable: `cases` and `population`.

A messy version

The same numbers, but the values `cases` and `population` are hidden in a `type` column:

```
1 table2
```

```
# A tibble: 12 × 4
```

	country	year	type	count
	<chr>	<dbl>	<chr>	<dbl>
1	Afghanistan	1999	cases	745
2	Afghanistan	1999	population	19987071
3	Afghanistan	2000	cases	2666
4	Afghanistan	2000	population	20595360
5	Brazil	1999	cases	37737
6	Brazil	1999	population	172006362
7	Brazil	2000	cases	80488
8	Brazil	2000	population	174504898
9	China	1999	cases	212258
10	China	1999	population	1270015070

Now a single observation (one country-year) is spread over **two rows**.